Project Report

On

Design and Implementation of an IR-Based Pulse Detector

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Abstract

This project presents the design and implementation of an infrared (IR)-based pulse detector using discrete electronic components and operational amplifiers. The system utilizes an infrared light-emitting diode (IR LED) as the optical source and a BPV10NF photodiode as the light receiver. When a finger is placed in the sensing region, variations in the received infrared light associated with pulsatile changes in blood volume produce a corresponding electrical signal. This signal is processed using LM358 operational amplifiers, reference-voltage circuitry, and additional passive components to improve its detectability. The conditioned signal drives an indicator LED, providing a visible indication of detected pulse-related variations.

The project involves circuit design, component selection, practical breadboard implementation, and experimental verification of the detector's operation. Particular attention is given to signal amplification, threshold detection, sensor positioning, and output stability. The completed circuit demonstrates the feasibility of detecting pulse-related optical variations using a relatively simple and low-cost analog electronic system. However, its sensitivity to finger placement and other environmental factors highlights the importance of appropriate sensor alignment and signal conditioning. The project provides practical experience in analog circuit design and demonstrates a potential foundation for future development of optical pulse-monitoring systems.

Keywords: Infrared sensing, pulse detection, BPV10NF photodiode, LM358, analog electronics, signal conditioning.

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1. Motivation

1.1 Background and Motivation

Heart rate is an important physiological parameter that provides useful information about cardiovascular activity. Pulse monitoring is widely used in healthcare, fitness monitoring, and biomedical instrumentation. The development of electronic pulse detection systems demonstrates how fundamental principles of electronics can be applied to detect physiological changes and convert them into measurable electrical signals.

This project focuses on developing a low-cost, infrared (IR)-based pulse detector using readily available electronic components. The system operates by directing infrared light toward a fingertip and detecting variations in the received light intensity associated with pulsatile changes in blood volume. A photodetector converts these variations into electrical signals, which are subsequently amplified and processed using operational amplifiers and a threshold detection stage. The resulting output is indicated by an LED, providing a visible representation of detected pulse activity.

The motivation behind this project is to gain practical experience in sensor interfacing, analog signal amplification, threshold detection, and biomedical signal processing. It also provides an opportunity to investigate real-world challenges such as noise interference, finger positioning, and signal stability. Through the construction and testing of the prototype, theoretical knowledge of electronic circuits can be applied to a practical problem while establishing a foundation for future improvements in pulse measurement and digital monitoring.

1.2 Conflicting Requirements

The design of an IR-based pulse detector involves several conflicting requirements that must be balanced to achieve reliable operation.

- **Sensitivity versus noise immunity:** The system must detect small variations in received infrared light while avoiding false detections caused by ambient light, electrical interference, and unwanted finger movement.
- **Amplification versus signal saturation:** Sufficient amplification is necessary to make weak photodetector signals detectable. However, excessive gain may cause the output to saturate, reducing the ability to distinguish individual pulses.
- **Detection reliability versus finger-position sensitivity:** The detector should respond consistently to changes in blood volume, but variations in finger placement, contact, and alignment can affect the received optical signal.
- **Low cost versus measurement stability:** Using inexpensive and readily available components helps keep the project affordable, although additional filtering, improved sensor arrangements, or more advanced processing may be required to improve stability.

- **Simple implementation versus measurement accuracy:** An analog circuit with LED indication is relatively straightforward to construct, but obtaining reliable numerical heart-rate measurements requires consistent pulse detection and appropriate timing or counting methods.

These trade-offs influence component selection, circuit configuration, amplification, threshold adjustment, and experimental testing. The design therefore aims to achieve a practical balance between simplicity, affordability, sensitivity, and reliable pulse detection.

1.3 Project Contribution

The main contribution of this project is the implementation of a functional IR-based pulse detector using readily available electronic components. In addition to detecting pulse-related variations and producing LED indications, the project incorporates an Arduino-based oscilloscope setup to observe and analyze the detector's electrical output. This measurement arrangement allows the output waveform and pulse periodicity to be investigated beyond simple visual observation of the LED. The project also explores practical issues such as finger-placement sensitivity, signal fluctuations, and the challenges of obtaining consistent pulse measurements from an analog detection circuit. These features provide a foundation for future improvements, including more stable pulse counting, digital heart-rate estimation, and microcontroller-based visualization. The prototype is intended for educational and experimental purposes rather than clinical diagnosis.

1.4 Report Organization

This report has been organized into eight main sections. Section 1, entitled “Motivation,” presents the background and motivation, conflicting design requirements, project contribution, and organization of the report. Section 2 describes the objectives of the project. Section 3, “Methodology and Implementation,” explains the operating principle, system flowchart, circuit connections, and theoretical background of the hardware and software components used. Section 4, “Final Outcome Analysis,” presents the simulation results, experimental observations, waveform analysis, and discussion of the system's performance, accuracy, and implementation challenges. Section 5 presents the work timeline using an appropriate project-planning chart. Section 6 provides the cost analysis of the project components and implementation. Section 7 discusses the project's impact on society and the environment. Finally, Section 8 presents the conclusions, summarizes the major achievements, and identifies potential future improvements. The report concludes with the relevant references used throughout the project.

2. Objectives

1. To develop an IR-based sensing system capable of detecting pulse-related variations in blood volume through a fingertip using an infrared transmitter and photodetector.
2. To design an analog signal-conditioning circuit using LM358 operational amplifiers to amplify the weak electrical signal generated by the photodetector.
3. To implement a threshold detection circuit that converts the conditioned signal into detectable output pulses for indicating pulse activity.
4. To provide visual indication of detected pulse activity through an LED and investigate the stability of the output under practical operating conditions.
5. To observe and analyze the detector's output waveform using an Arduino-based oscilloscope setup and examine the time intervals between successive detected pulses.
6. To evaluate the performance and limitations of the prototype, estimate the pulse rate in beats per minute (BPM) where reliable pulse detection is possible, and identify potential improvements in detection stability and measurement accuracy.

3. Methodology and Implementation

3.1 Working Principle of Optical Pulse Detection

The proposed pulse detector operates on the principle of optical sensing, in which variations in light intensity are used to detect pulse-related changes in blood volume. The system consists of an infrared (IR) light-emitting diode (LED), a photodetector, signal amplification stages, a threshold detection circuit, and an LED indicator. When a fingertip is positioned in the sensing region, the received infrared light varies with changes in the volume of blood associated with each cardiac cycle. These variations form the basis of pulse detection.

The IR LED acts as the optical transmitter, directing infrared light toward the fingertip. As blood volume changes during successive cardiac cycles, the absorption and scattering of the transmitted light also vary. The photodetector senses the resulting changes in received light intensity and converts them into a corresponding electrical signal. Since these variations are relatively small, the resulting signal requires amplification before it can be processed reliably.

The electrical signal from the photodetector is processed using LM358 operational amplifiers. The amplification stages increase the signal amplitude so that pulse-related variations can be distinguished from the surrounding signal level. The conditioned signal is then applied to a threshold detection stage, which determines whether the signal has crossed a specified voltage level. When the threshold condition is satisfied, the circuit produces an output transition that controls the LED indicator. Consequently, changes in the electrical output provide a visible indication of detected pulse activity.

For experimental observation, the detector's electrical output is also connected to an Arduino-based oscilloscope setup. The Arduino samples the selected measurement point and transmits the acquired data to a computer for waveform visualization. By examining successive output events, the periodic behavior of the detector can be investigated, and the time interval between detected pulses can be estimated. If individual pulse events are detected consistently, the pulse rate can be estimated from the measured period using the relationship between frequency and time.

The overall operation therefore involves optical transmission, photodetection, signal amplification, threshold-based detection, and visual indication. The reliability of the detected output depends on factors such as finger placement, ambient light, motion artifacts, signal amplification, and threshold adjustment. Since the circuit detects pulse-related optical variations rather than directly measuring blood pressure or identifying the mechanical phases of the cardiac cycle, its primary function is pulse activity detection rather than clinical cardiovascular assessment.

3.2 Flowchart of System Operation

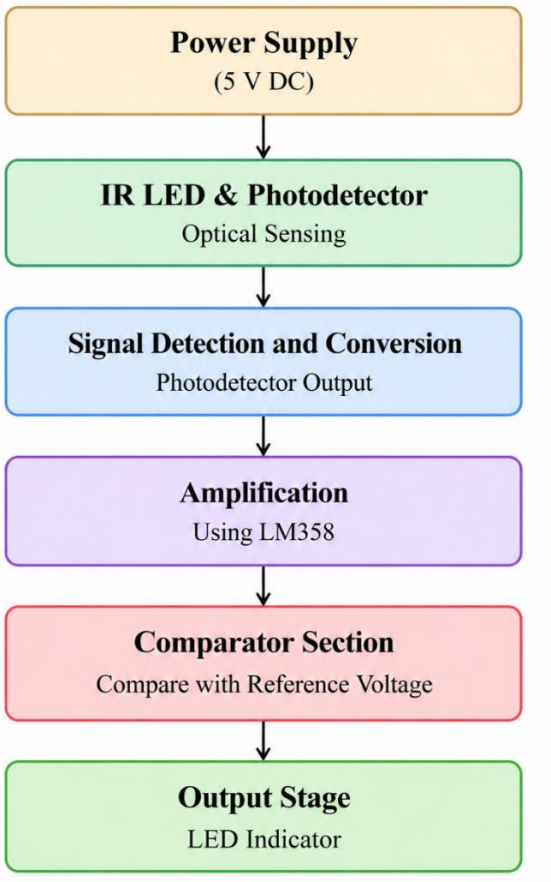


Fig-3.1: Flowchart of System Operation

3.3 Description of Each Block

1. Power Supply

The power supply provides the necessary DC voltage to operate the electronic components of the pulse detector. It supplies electrical energy to the IR LED, photodetector circuit, LM358 operational amplifiers, comparator section, and output stage according to their respective voltage requirements. A stable power supply is essential for consistent circuit operation.

2. IR LED and Photodetector

The IR LED acts as the infrared light source and directs light toward the fingertip. The photodetector receives the transmitted or reflected infrared light, depending on the sensor arrangement. Changes in the received light intensity associated with pulsatile variations in blood volume produce corresponding changes in the photodetector's electrical response.

3. Signal Detection and Conversion

The photodetector converts variations in received infrared light into an electrical signal. This signal contains pulse-related variations along with a background component and possible noise. The resulting electrical signal serves as the input to the amplification stage for further processing.

4. Amplification Using LM358

The LM358 operational amplifier is used to amplify the weak electrical signal obtained from the photodetector. Amplification increases the magnitude of the pulse-related variations, making them easier to distinguish and process in the subsequent comparator stage. Appropriate gain is important to avoid excessive amplification and output saturation.

5. Comparator Section

The comparator section compares the conditioned signal with a predefined reference voltage. When the signal crosses the reference level, the comparator output changes state, generating an electrical transition corresponding to a detected pulse event. The reference voltage determines the detection threshold and influences the circuit's sensitivity to pulse-related variations.

6. Output Stage

The output stage provides a visible indication of detected pulse activity through an LED. The LED responds to changes in the comparator output, allowing pulse events to be observed without requiring a separate display. The timing of successive output events can also be examined during experimental testing to investigate pulse periodicity and estimate pulse rate.

3.4 Circuit Design and Connection Diagram

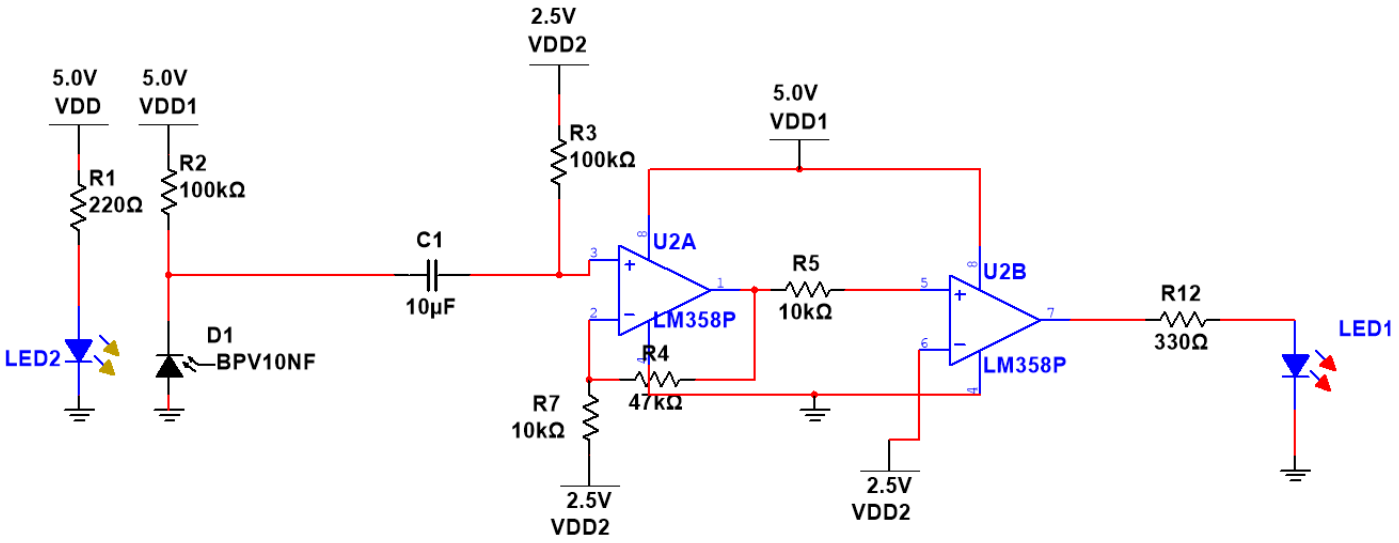


Fig-3.2: Circuit Diagram of the Heartbeat Detector (Simulated)

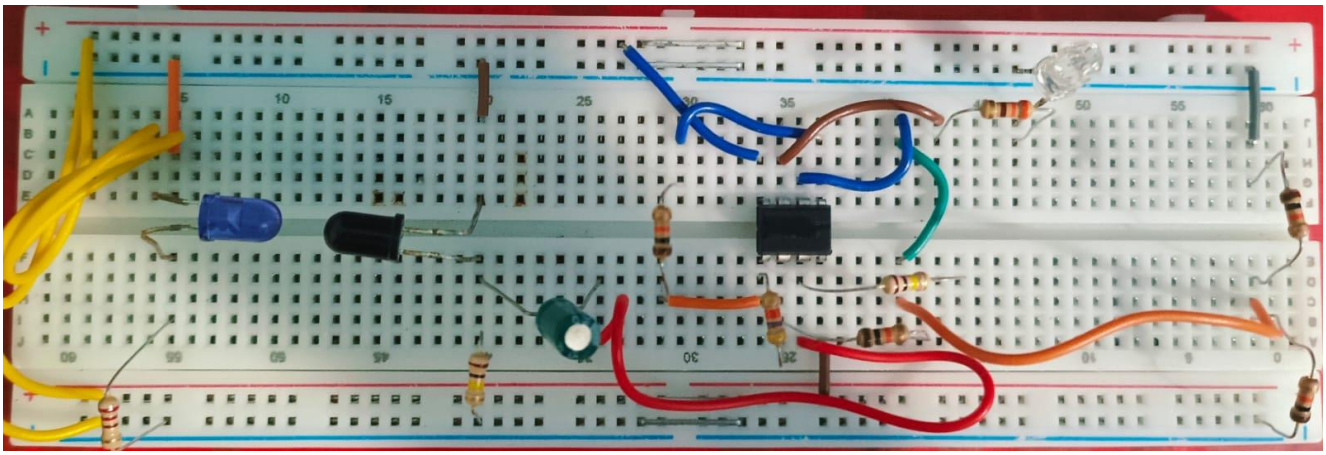


Fig-3.3: Practical Implementation of the Heartbeat Detector (Real-life)

3.5 Apparatus Required

Table 3.1: List of Components and Apparatus

Sl. No.	Component / Apparatus	Quantity	Function
1	Infrared (IR) LED	1	Emits infrared light toward the fingertip.
2	Photodetector (BPV10NF)	1	Detects variations in received infrared light.
3	LM358 operational amplifier IC	1	Amplifies the photodetector signal and supports comparator operation as configured.
4	LED indicator	1	Provides visible indication of detected pulse activity.
5	Resistors	7	Set operating currents, amplifier gain, and reference voltage.
6	Capacitors	1	Provide circuit functions such as coupling or stabilization were included in the actual design.
7	DC power supply	1	Supplies the required operating voltage to the circuit.
8	Breadboard	1	Supports circuit assembly and prototyping.
9	Connecting wires	As required	Establish electrical connections between components.

3.6 Theoretical Background of Components

1. Resistor

A resistor is a passive electronic component that opposes the flow of electric current. It is used to limit current and establish voltage levels in electronic circuits. Its operation follows Ohm's law:

$$V = IR$$

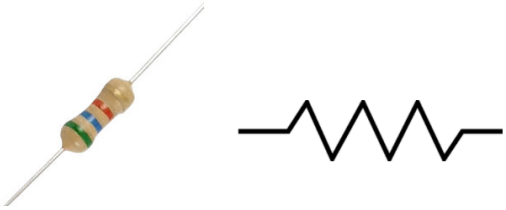


Fig-3.4: Resistor and It's Symbol

2. Capacitor

A capacitor is a passive component that stores electrical energy in an electric field between two conductive plates. It can store and release charge and is commonly used for coupling, decoupling, and filtering applications.

$$Q = CV$$

where (Q) is charge, (C) is capacitance, and (V) is voltage.

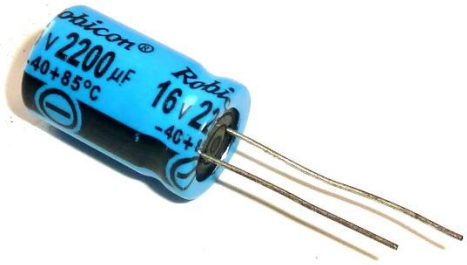


Fig-3.5: Capacitor (Electrolytic)

3. Regular Light-Emitting Diode (LED)

A regular LED is a semiconductor diode that emits visible light when forward biased. It converts electrical energy into light energy and is commonly used as an indicator in electronic circuits.



Fig-3.6: LED (RED)

4. Infrared Light-Emitting Diode (IR LED)

An IR LED is a semiconductor diode that emits infrared radiation when forward biased. Unlike a regular LED, it produces light that is generally invisible to the human eye. It is commonly used in optical sensing, remote controls, and infrared communication systems.



Fig-3.7: IR LED

5. Photodiode

A photodiode is a semiconductor device that converts incident light into electrical current. Its photocurrent varies with the intensity of received light, making it useful in optical sensing and light-detection applications.



Fig-3.8: Photodetector (BPV10NF)

6. Operational Amplifier

An operational amplifier is a high-gain electronic amplifier with inverting and non-inverting inputs. With suitable external components and feedback, it can perform signal amplification, comparison, and other analog operations.

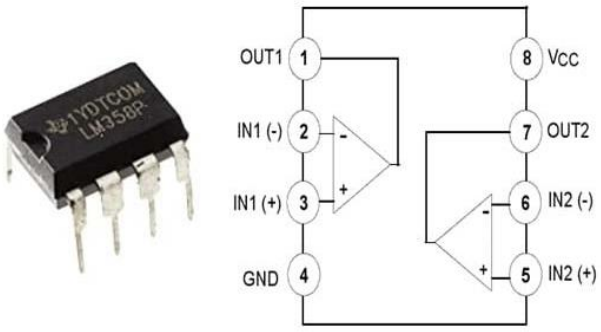


Fig-3.9: Op-Amp (LM-358)

4. Final Outcome Analysis

4.1 Simulations

The pulse detector circuit was simulated using NI Multisim to examine its response to a periodic input signal. Since the optical behavior of the IR LED, fingertip, and photodetector was not modeled directly, a pulse voltage source was used to provide a controlled electrical input to the signal-conditioning circuit.

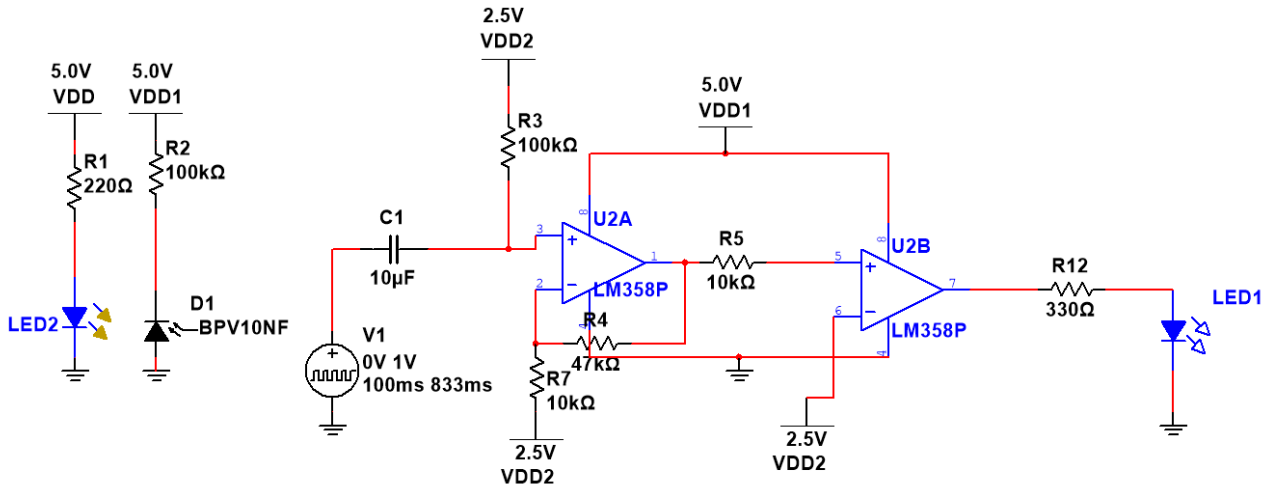


Fig-4.1: Multisim circuit diagram of the pulse detector.

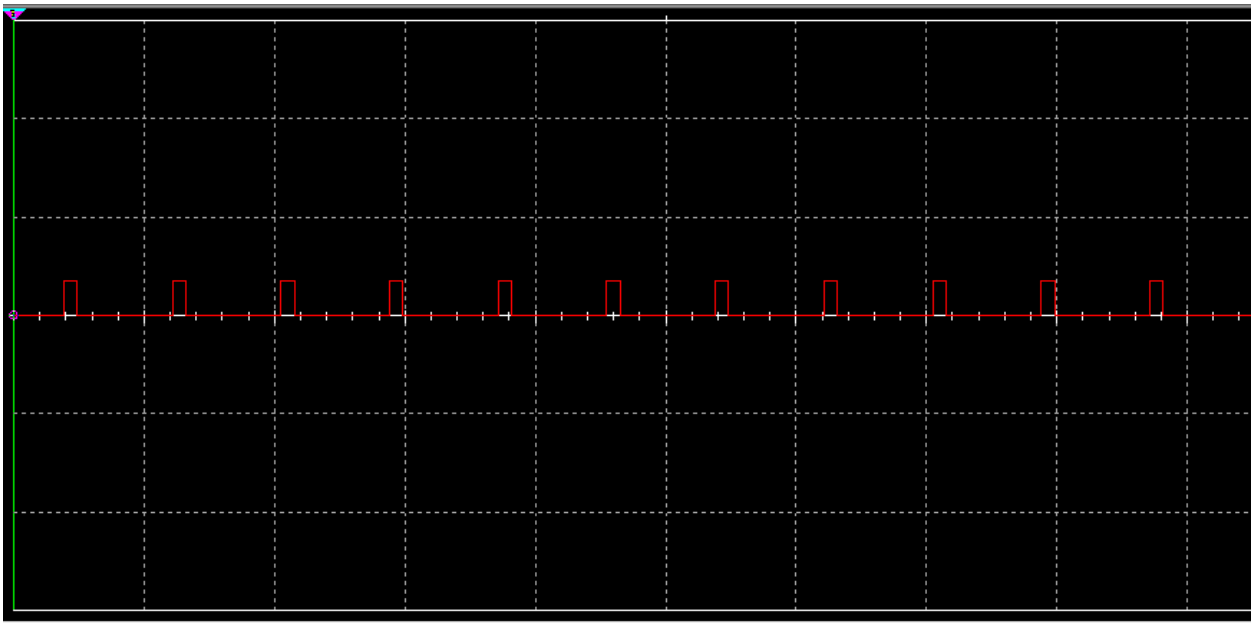


Fig-4.2: Simulated output waveform observed using the virtual oscilloscope.

4.1.1 Input Signal and Parameters

Table-4.1: The pulse voltage source was configured with the following parameters:

Parameter	**Value**
Initial voltage, (V_1)	0 V
Pulsed voltage, (V_2)	1 V
Pulse width, (T_{on})	100 ms
Period, (T)	833 ms
Rise time	1 ns
Fall time	1 ns

The pulse source was used to simulate repetitive electrical variations similar to the changes that the photodetector signal may undergo during pulse sensing. This allowed the amplifier, comparator, and LED output stages to be tested under controlled conditions.

4.1.2 Mathematical Analysis

The pulse repetition frequency is calculated from the period:

$$\left[f = \frac{1}{T}\right]$$

Given,

$$[T = 833 \text{ ms} = 0.833 \text{ s}]$$

Therefore,

$$\left[f = \frac{1}{0.833} \approx 1.2005 \text{ Hz}\right]$$

The corresponding simulated pulse rate is

$$[\text{Pulse Rate} = 60f]$$

$$[= 60(1.2005) \approx 72.03]$$

Thus, the source generates approximately **72 pulses per minute**.

4.1.3 Simulation Observation

The virtual oscilloscope displayed a periodic rectangular waveform with regularly spaced pulses. This observation confirmed the repetitive nature of the applied input signal. The circuit was also used to examine the response of the LM358 amplification stage, comparator section, and LED indicator to the input pulses.

The simulation demonstrates the response of the electronic circuit to a controlled input. However, the calculated rate of approximately 72 pulses per minute represents the **simulated input repetition rate**, not an actual human heart-rate measurement. The pulse source also does not reproduce the complete analog behavior of the photodetector or the optical interaction with biological tissue.

4.2 Result Analysis

The output waveforms of the IR-based pulse detector were recorded to evaluate the consistency of pulse detection and estimate the pulse rate. The measured voltage transitions represent the output states of the detection circuit. Three waveform records are presented to examine the regularity of the detected events and the stability of the system.

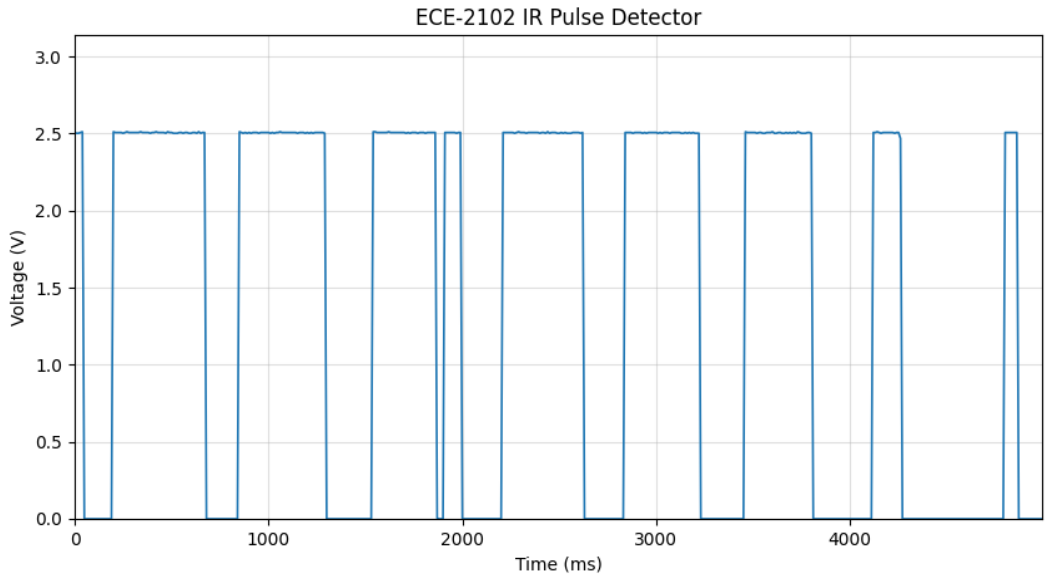


Fig-4.3: First recorded output waveform of the IR-based pulse detector.

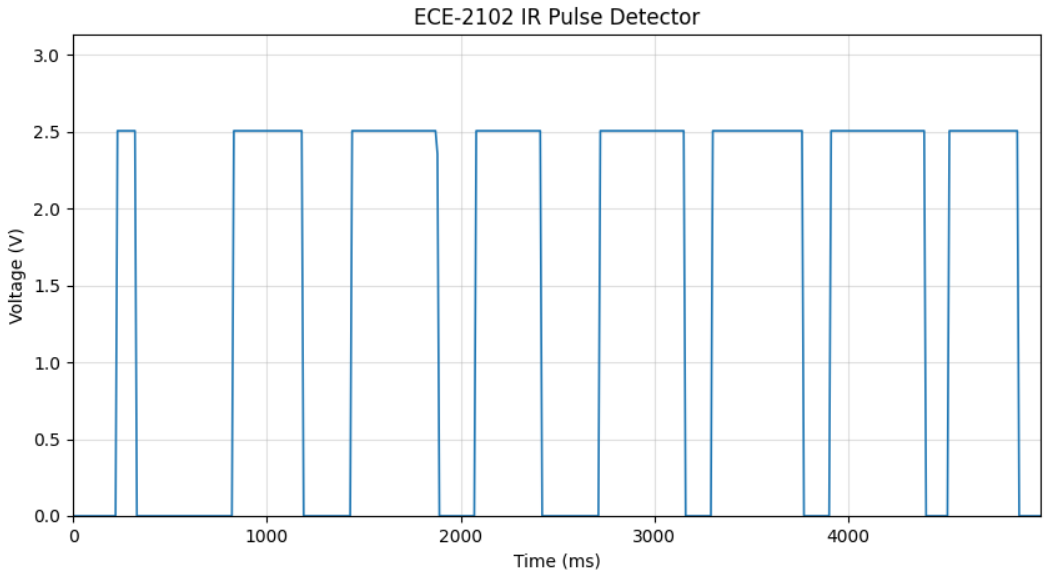


Fig-4.4: Second recorded output waveform showing relatively regular pulse events.

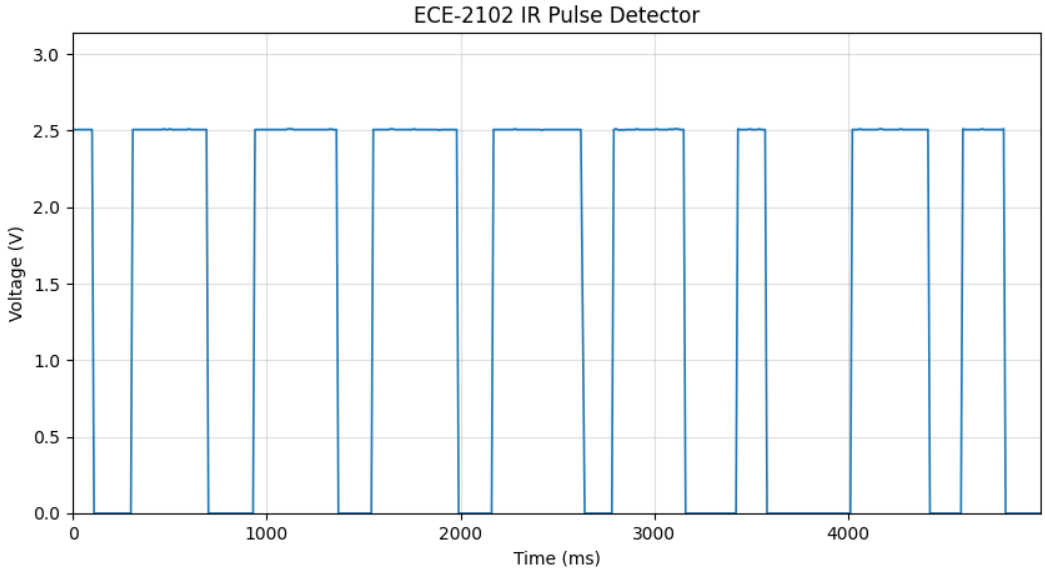


Fig-4.5: Third recorded output waveform illustrating variations in pulse detection.

4.2.1 Pulse Rate Calculation

The pulse rate can be estimated by measuring the time interval between successive detected pulse events. If (T) is the average time between consecutive pulses in seconds, the pulse rate is calculated as

$$\boxed{\text{BPM} = \frac{60}{T}}$$

For the second waveform, the time axis extends to approximately $(5000,\text{ms})$. The regularly spaced pulse events have an approximate average separation of $(0.61,\text{s})$, based on visual estimation from the plotted waveform.

Therefore,

$$T \approx 0.61\text{s}$$

$$\text{BPM} = \frac{60}{0.61}$$

$$\boxed{\text{BPM} \approx 98.4}$$

Thus, the second waveform corresponds to an estimated detection rate of approximately **98 pulses per minute**. This is an estimate from the displayed graph rather than a precise measurement from the original sampled data.

4.2.2 Waveform Analysis

The three recorded waveforms show variations in the timing and consistency of the detector's output pulses. The second waveform exhibits relatively regular pulse intervals, while the first and third waveforms contain irregular intervals and occasional closely spaced transitions. These observations indicate variations in pulse detection consistency.

4.3 Discussion

The experiment demonstrated the operation of an IR-based pulse detector using optical sensing, LM358 amplification, comparator-based detection, and LED indication. Multisim simulation helped verify the circuit's response to periodic input signals, while experimental waveform observations provided insight into its practical performance.

The recorded waveforms showed variations in pulse detection consistency, potentially caused by finger movement, sensor positioning, ambient light, and threshold settings. The second waveform suggested an estimated pulse rate of approximately 98 BPM, although this value requires verification using the original sampled data.

Overall, the project successfully demonstrated the basic principle of electronic pulse detection. Further improvements in signal stability and comparison with a reference heart-rate monitor are needed to evaluate measurement accuracy and reliability.

5. Work Timeline

The project was organized into three weeks of activities, covering initial planning, circuit design, hardware implementation, simulation, experimental testing, and report preparation. The tasks were arranged to facilitate systematic development and evaluation of the IR-based pulse detector. Some activities overlapped to allow iterative testing and troubleshooting.

Table 5.1: Gantt Chart of the Project Work Timeline.

Project activity	Week 1	Week 2	Week 3
Project planning and component selection	■		
Circuit design and theoretical calculations	■		
Hardware construction	■	■	
Multisim simulation		■	
Circuit testing and troubleshooting		■	■
Oscilloscope measurements and waveform analysis			■
Report writing and final review		■	■

6. Cost Analysis

Table 6.1: Table for the expenses for this project

Sl.	Item	Justification	Price (BDT)
1	IR LED and BPV10NF photodiode	Infrared transmission and pulse sensing	20
2	2 × LM358P ICs	Signal amplification and pulse detection + 1 backup IC	40
3	Resistors and capacitors	Biasing, amplification, and signal conditioning	30
4	Breadboard and Wiring Kit	Circuit assembly and interconnection	300
5	LED and miscellaneous components	Visual indication and supporting connections	20
Total (BDT): 410 BDT			
In words: Four hundred and ten taka only.			

7. Impact of Project on Society, Environment and Sustainability

Societal Impact: The project demonstrates an affordable method of detecting pulse-related changes using infrared technology and basic electronic components. It promotes awareness of optical pulse monitoring and provides practical knowledge of biomedical instrumentation. The prototype can serve as an educational tool for students studying analog electronics, sensor technology, and biomedical applications. Furthermore, the project illustrates how fundamental electronic principles can be applied to develop simple health-monitoring prototypes at a relatively low cost. However, the system is intended for educational purposes and is not suitable for medical diagnosis or clinical monitoring.

Environmental Impact: The circuit operates from a low-voltage DC supply and uses a relatively small number of electronic components, resulting in modest resource requirements during operation. Reusing breadboards, connecting wires, and laboratory equipment can reduce material consumption and electronic waste. The low-power design also helps limit energy consumption. Proper maintenance, component reuse, and responsible disposal of damaged electronic parts can further minimize the environmental impact.

Sustainability: The use of inexpensive and readily available components makes the project economical to maintain, repair, and modify. Individual components can be replaced when necessary, extending the useful life of the prototype and reducing the need to discard the entire circuit. The design can also serve as a foundation for future improvements, including better sensor positioning, more stable pulse detection, and automatic pulse-rate calculation. These possibilities support continued technical learning, affordable prototyping, and the sustainable development of basic electronic instrumentation.

8. Conclusions

The project successfully demonstrated the basic operation of an IR-based pulse detector using optical sensing, LM358 amplification, comparator-based detection, and LED indication. Multisim simulation helped examine the circuit's response to periodic input signals, while experimental testing and Arduino-based oscilloscope measurements provided useful observations of the output waveforms. The results confirmed that the circuit could produce detectable pulses, although variations in waveform stability were observed during practical testing.

The project provided practical experience in analog circuit design, component selection, simulation, and experimental waveform analysis. However, factors such as finger positioning, movement, ambient light, and threshold settings may affect detection consistency. Future improvements could include better signal conditioning, automatic BPM calculation, and comparison with a reference heart-rate monitor to evaluate accuracy and reliability. Overall, the project achieved its primary objective of demonstrating the fundamental principle of electronic pulse detection.

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